

Code

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1a.

```
#load in the many libraries
library(ggplot2)
library(dplyr)
library(matrixStats)
library(gt)
library(gtExtras)
# Make results reproducible
set.seed(5771)

#NOTE: I just reused this for every sample size and changed the title in the ggplot

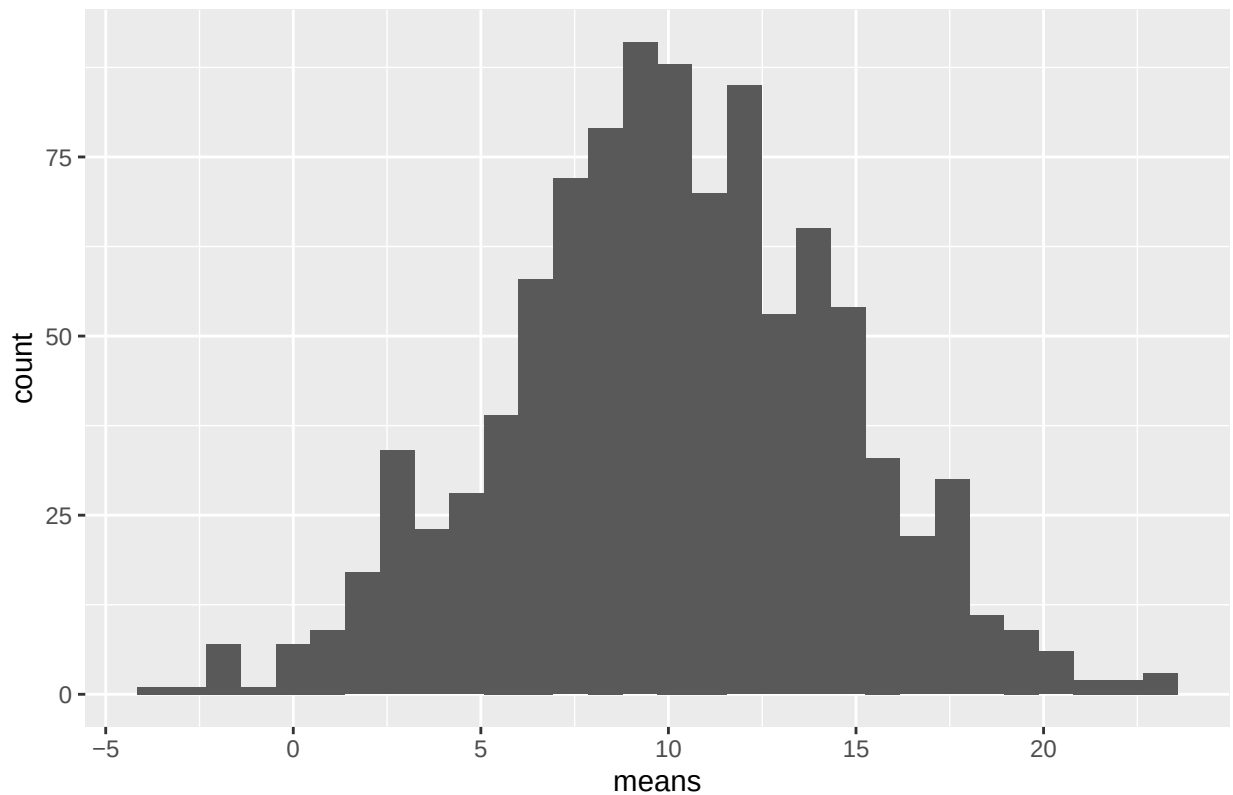
# Define sample size
n <- 5 # you'll also try 10, 30, 100, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 1: normal
#getting the data
normal_data <- replicate(1000, rnorm(n, mean = mu, sd = sigma))

#taking the means and turning into a df
normal_means <- data.frame(colMeans(normal_data))
#changing the name of the column
colnames(normal_means)[1] <- "means"
#finding the sd of
sd <- colSds(normal_data)
#adding the sd to our means df
normal_means <- normal_means %>%
  cbind(colSds(normal_data))
#rename the column
colnames(normal_means)[2] <- "sd"

#graphing
ggplot(data = normal_means) + geom_histogram(aes(x = means)) +
  labs(title = "Histogram of Means for 1000 Samples: Size 5")
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Histogram of Means for 1000 Samples: Size 5



1b.

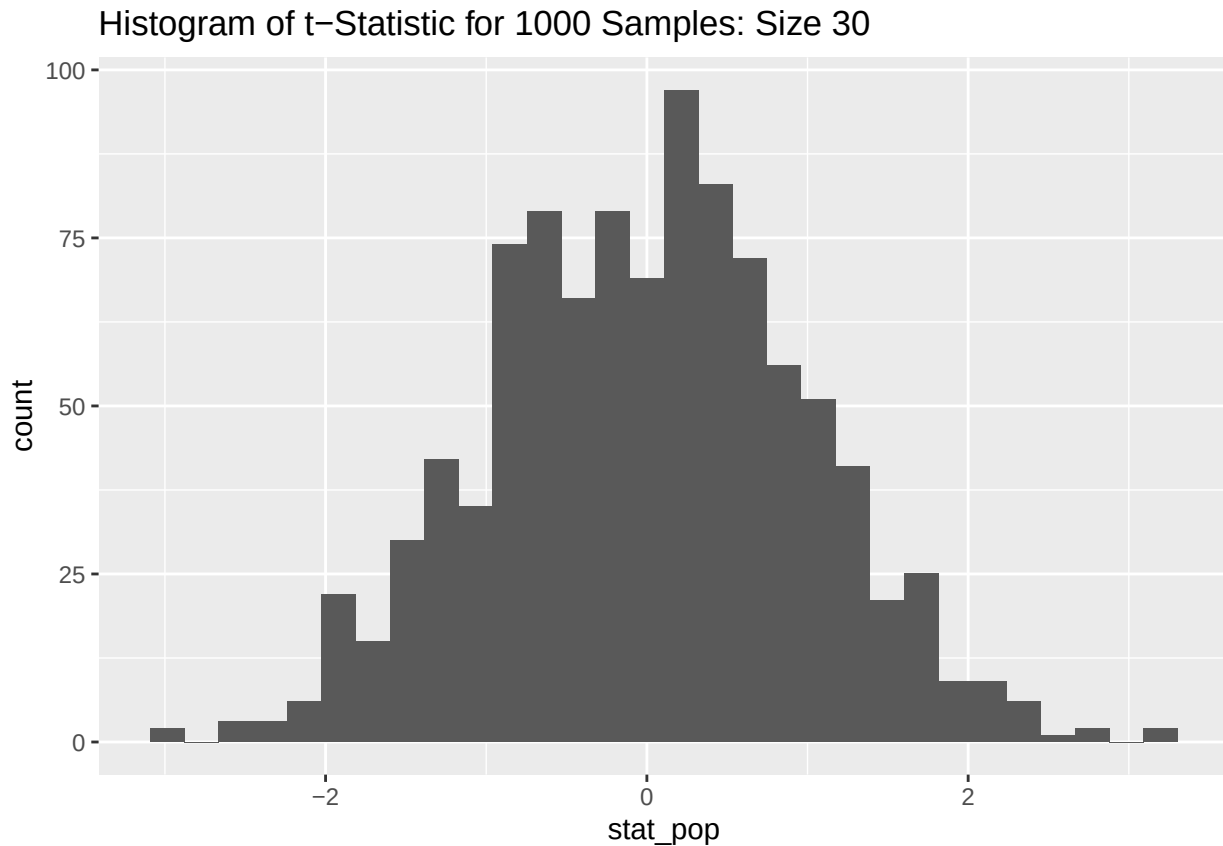
```
n <- 5 # you'll also try 10, 30, 100, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 1: normal
#getting the data
normal_data <- replicate(1000, rnorm(n, mean = mu, sd = sigma))

#taking the means and turning into a df
normal_means <- data.frame(colMeans(normal_data))
#changing the name of the column
colnames(normal_means)[1] <- "means"
#finding the sd of
sd <- colSds(normal_data)
#adding the sd to our means df
normal_means <- normal_means %>%
  cbind(colSds(normal_data))
#rename the column
colnames(normal_means)[2] <- "sd"

#calculating the t statistic with the pop sd
normal_means <- normal_means %>%
  mutate(stat_pop = ((means - 10)/(10/sqrt(n))))
```

```
#graph
ggplot(data = normal_means) + geom_histogram(aes(x = stat_pop)) +
  labs(title = "Histogram of t-Statistic for 1000 Samples: Size 30")
```

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.



1c.

```
n <- 500 # you'll also try 10, 30, 100, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 1: normal
#getting the data
normal_data <- replicate(1000, rnorm(n, mean = mu, sd = sigma))

#taking the means and turning into a df
normal_means <- data.frame(colMeans(normal_data))
#changing the name of the column
colnames(normal_means)[1] <- "means"
#finding the sd of
sd <- colSds(normal_data)
#adding the sd to our means df
normal_means <- normal_means %>%
  cbind(colSds(normal_data))
```

```

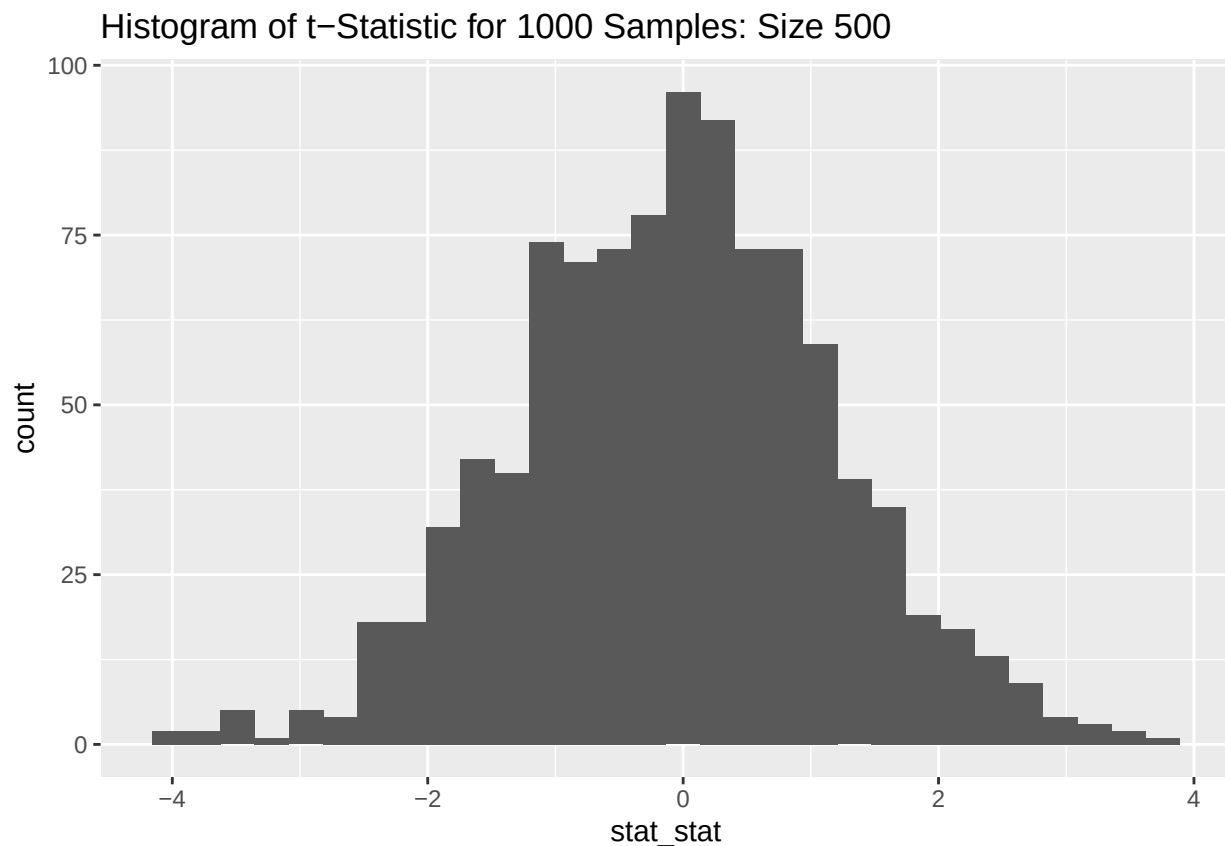
#rename the column
colnames(normal_means)[2] <- "sd"

#calculating the t statistic with the pop sd
normal_means <- normal_means %>%
  mutate(stat_stat = ((means - sd)/(10/sqrt(n))))

#graph
ggplot(data = normal_means) + geom_histogram(aes(x = stat_stat)) +
  labs(title = "Histogram of t-Statistic for 1000 Samples: Size 500")

```

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.



d.

```

n <- 500 # you'll also try 10, 30, 100, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 1: normal
#getting the data
normal_data <- replicate(1000, rnorm(n, mean = mu, sd = sigma))

#taking the means and turning into a df

```

```

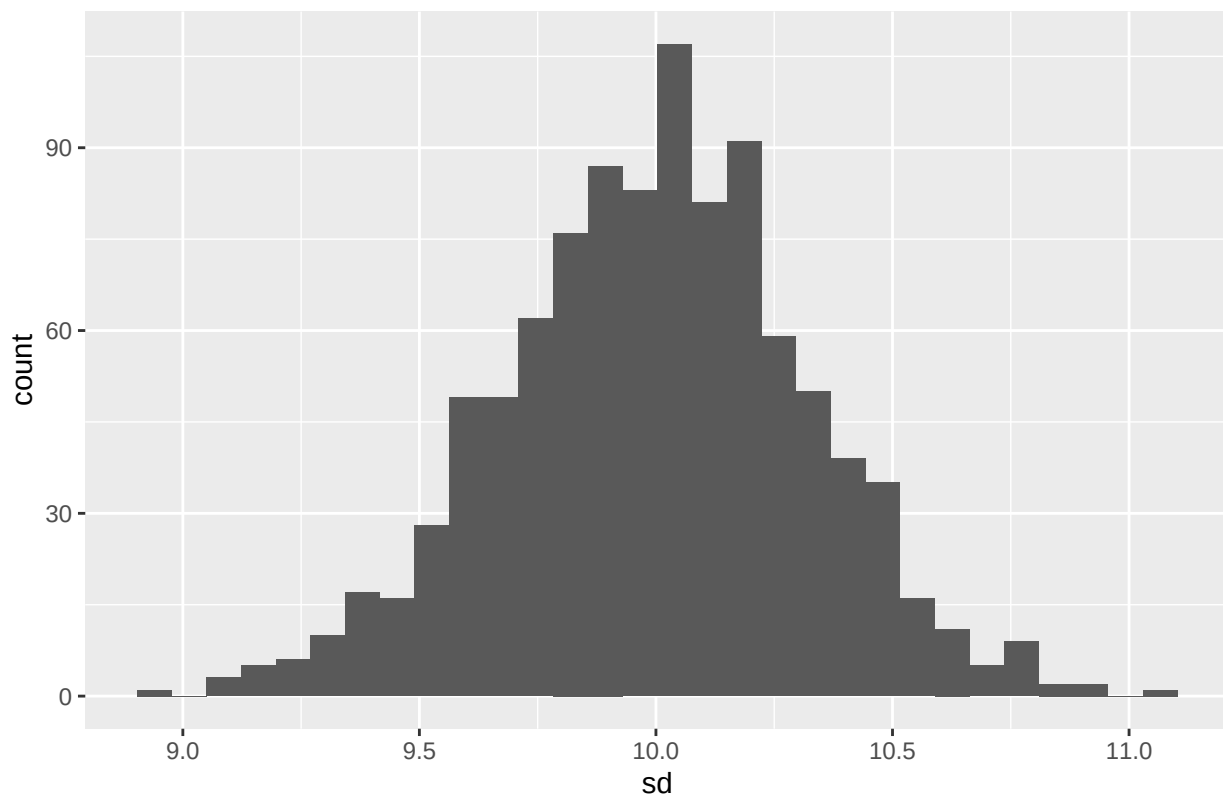
normal_means <- data.frame(colMeans(normal_data))
#changing the name of the column
colnames(normal_means)[1] <- "means"
#finding the sd of
sd <- colSds(normal_data)
#adding the sd to our means df
normal_means <- normal_means %>%
  cbind(colSds(normal_data))
#rename the column
colnames(normal_means)[2] <- "sd"

#graph
ggplot(data = normal_means) + geom_histogram(aes(x = sd)) +
  labs(title = "Histogram of SD for 1000 Samples: Size 500")

```

'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.

Histogram of SD for 1000 Samples: Size 500



e.

```

n <- 30 # you'll also try 10, 30, 100, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 1: normal
#getting the data

```

```

normal_data <- replicate(1000, rnorm(n, mean = mu, sd = sigma))

#taking the means and turning into a df
normal_means <- data.frame(colMeans(normal_data))
#changing the name of the column
colnames(normal_means)[1] <- "means"
#finding the sd of
sd <- colSds(normal_data)
#adding the sd to our means df
normal_means <- normal_means %>%
  cbind(colSds(normal_data))
#rename the column
colnames(normal_means)[2] <- "sd"

#looks fine
#calculating the test statistic with the s and n!
normal_means_sd <- normal_means %>%
  mutate(stat_pop = ((means - mu)/(sd/sqrt(n))))

#count how many are greater than 1.96 and smaller than -1.96
p_value <- normal_means_sd %>%
  filter(stat_pop > 1.96 | stat_pop < -1.96)
#117 observations are stat sig for 5
#80 are stat sig for 10
#50 are stat sig for 30
#47 are stat sig for 100
#53 are stat sig for 500

#calculating the t-statistic with the population and N
normal_means_pop <- normal_means %>%
  mutate(stat_pop = ((means - mu)/(sigma/sqrt(n))))
p_value <- normal_means_pop %>%
  filter(stat_pop > 1.96 | stat_pop < -1.96)
#get 47 for stat sig for 5
#get 60 for stat sig for 10
#get 48 for 30
#41 are stat sig for 100
#57 are stat sig for 500

#using a t-statistic with the sd from the groups and a t-dist
p_value_tdist <- normal_means_sd %>%
  filter(stat_pop > 1.699 | stat_pop < -1.833)
#98 are stat sig for for 5
#102 are stat sig for 10
#33 are stat sig for 30
#31 are stat sig for 100
#36 are stat sig for 500

```

le.

```

#pretty table time LETS GOOOO
names <- NA
n_5 <- NA
n_10 <- NA
n_30 <- NA
n_100 <- NA
n_500 <- NA

table <- data.frame(names, n_5, n_10, n_30, n_100, n_500)
table[1, ] <- list("Populatin SD", "47", "60", "48", "41", "57")
table[nrow(table) + 1, ] <- list("Sample SD", "117", "80", "50", "47", "53")
table[nrow(table) + 1, ] <- list("t-distribution and Sample SD", "98", "102", "33", "31", "36")

gt_table <- gt(table)
gt_table <- gt_table %>%
  cols_label(
    names = "Sample Size:",
    n_5 = "5",
    n_10 = "10",
    n_30 = "30",
    n_100 = "100",
    n_500 = "500"
  ) %>%
  gt_add_divider(columns = "names")

gt_table

```

Sample Size:	5	10	30	100	500
Populatin SD	47	60	48	41	57
Sample SD	117	80	50	47	53
t-distribution and Sample SD	98	102	33	31	36

```

names <- NA
n_5 <- NA
n_10 <- NA
n_30 <- NA
n_100 <- NA
n_500 <- NA

table <- data.frame(names, n_5, n_10, n_30, n_100, n_500)
table[1, ] <- list("df", "4", "9", "29", "99", "499")
table[nrow(table) + 1, ] <- list("Critical Value", "2.132", "1.833", "1.699", "1.660", "1.648")

gt_table <- gt(table)
gt_table <- gt_table %>%
  cols_label(
    names = "Sample Size:",
    n_5 = "5",
    n_10 = "10",
    n_30 = "30",

```

```

n_100 = "100",
n_500 = "500"
) %>%
gt_add_divider(columns = "names")

gt_table

```

Sample Size:	5	10	30	100	500
df	4	9	29	99	499
Critical Value	2.132	1.833	1.699	1.660	1.648

Part 2

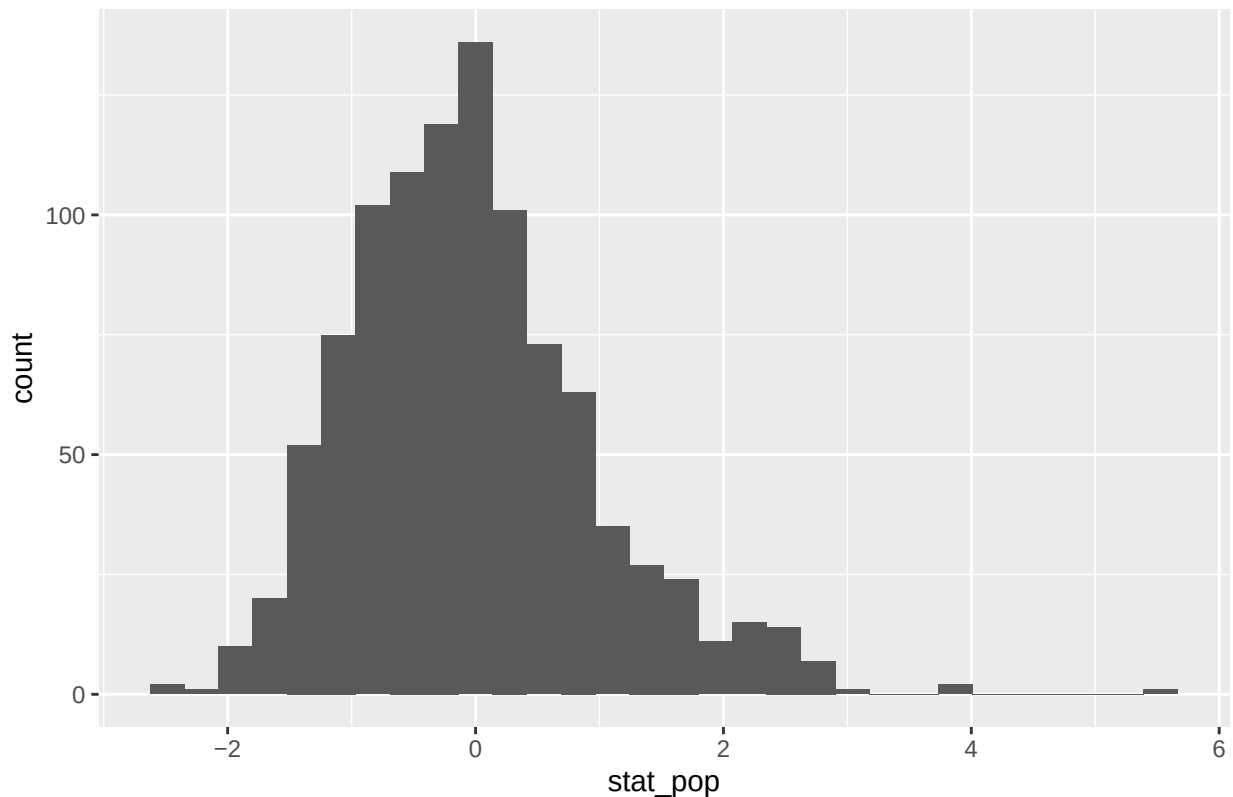
```

# Define sample size
n <- 10 # you'll only try 10, 30, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 2: gamma
gamma_data <- replicate(1000, rgamma(n, shape = mu^2/sigma^2, scale = sigma^2/mu))
#taking the means and turning into a df
gammadata <- data.frame(colMeans(gamma_data))
#changing the name of the column
colnames(gammadata)[1] <- "means"
#finding the sd of
sd <- colSds(gamma_data)
#adding the sd to our means df
gammadata <- gammadata %>%
  cbind(colSds(gamma_data))
#rename the column
colnames(gammadata)[2] <- "sd"
gammadata <- gammadata %>%
  mutate(stat_pop = ((means - 10)/(10/sqrt(n))))
#graph
ggplot(data = gammadata) + geom_histogram(aes(x = stat_pop)) +
  labs(title = "Gamma Distribution Test Statistic: Sample Size 500")

```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```


Gamma Distribution Test Statistic: Sample Size 500

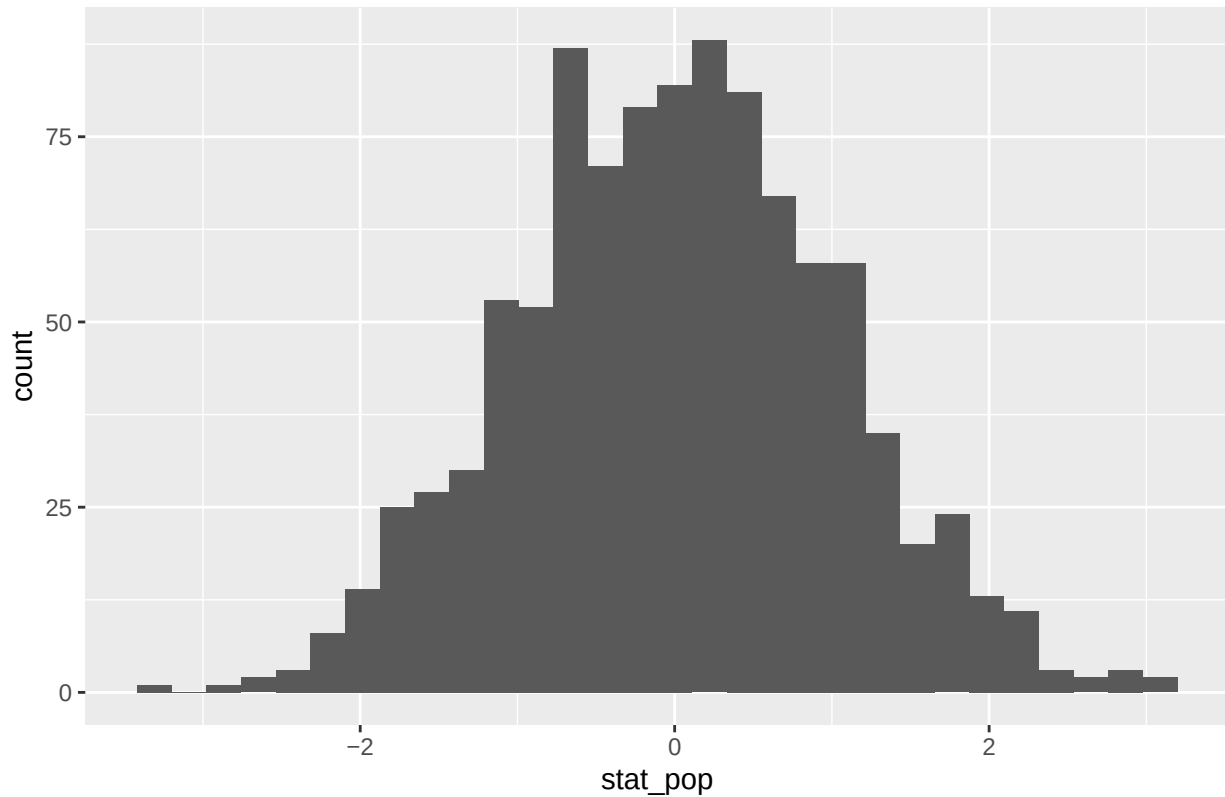


```
# Population 3: using the uniform, but we need them to have the same mu and sigma

n <- 10 # you'll only try 10, 30, and 500
# Simulate datasets from different populations with the same mean and standard deviation
mu <- 10
sigma <- 10
# Population 2: gamma
#so i calculated the needed min and max to get 10 and 10
data3 <- replicate(1000, runif(n, min= -7.32, max =27.32))
#taking the means and turning into a df
uniformdata <- data.frame(colMeans(data3))
#changing the name of the column
colnames(uniformdata)[1] <- "means"
#finding the sd of
sd <- colSds(data3)
#adding the sd to our means df
uniformdata <- uniformdata %>%
  cbind(colSds(data3))
#rename the column
colnames(uniformdata)[2] <- "sd"
uniformdata <- uniformdata %>%
  mutate(stat_pop = ((means - 10)/(10/sqrt(n))))
#graph
ggplot(data = uniformdata) + geom_histogram(aes(x = stat_pop)) +
  labs(title = "Uniform Distribution Test Statistic: Sample Size 10")
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value with 'binwidth'.
```

Uniform Distribution Test Statistic: Sample Size 10



```
passes_popsd <- uniformdata %>%  
  #take abs bc it has to be 1.96 away in either direction  
  mutate(teststatistic = abs(stat_pop))  
  #count how many are greater than 1.96  
nrow(passes_popsd[passes_popsd$teststatistic>1.648, ])
```

```
## [1] 112
```

```
#pretty table time LETS GOOOO  
names <- NA  
n_10 <- NA  
n_30 <- NA  
n_500 <- NA  
  
table <- data.frame(names, n_10, n_30, n_500)  
table[1, ] <- list("Normal Distribution", "63", "52", "48")  
table[nrow(table) + 1, ] <- list("Gamma Distribution", "98", "72", "51")  
table[nrow(table) + 1, ] <- list("Uniform Distribution", "101", "81", "55")  
  
gt_table <- gt(table)  
gt_table <- gt_table %>%  
  cols_label(  
    names = "Sample Size:",
```

```
n_10 = "10",
n_30 = "30",
n_500 = "500"
) %>%
gt_add_divider(columns = "names")

gt_table
```

Sample Size:	10	30	500
Normal Distribution	63	52	48
Gamma Distribution	98	72	51
Uniform Distribution	101	81	55